

01 / Spine measurement record

Collect nonloaded measurements from actual stock or an existing layout. Mark planned or unavailable dimensions as unresolved. The vessel supplies no structural bracing or string-load support.

Use inches, lbf and psi throughout; record instrument resolution and one-sided bounds. Origin: gross spine centroid in plan. x runs across the strings, y toward the strings, z upward along the spine. Mark the front face and a physical z datum.

Field	Measurement / input	Value	Unit	Evidence / bounds / method
M01	Actual configuration / drawing revision		text	
M02	Measurement recorder		text	
M03	Spine specimen / stock ID		text	
M04	Species		text	
M05	Grade or approved ungraded-member basis		text	
M06	Orientation / axes photo or drawing ID		text	
M07	Actual minimum width b across x		in	
M08	Actual minimum depth d toward strings		in	
M09	Overall actual spine length		in	
M10	Upper load-transfer reference z		in	
M11	Lower load-transfer reference z		in	
M12	Unsupported length for bending about x		in	
M13	Unsupported length for bending about y		in	
M18	Measured moisture content		%	
M19	Initial side bow toward x (magnitude)		in	
M20	Initial front/back bow toward y (magnitude)		in	
M21	Knots, checks, splits, grain slope, holes/notches		text	
M22	Design-property source and all adjustments		text	
M23	Existing stock / final dimensions evidence status		text	
M30	Measurement date		date	

Take width/depth readings at both joints and mid-span. Retain all readings and photos; enter conservative actual dimensions in the workbook. Do not treat a nominal 2 x 3, a selected part specification, or a proposed finished speaking length as an actual measurement.

02 / String measurements and limits

Companion tab: String Schedule. EX-B1 remains a source example; the actual product, scale, pitch and reviewed limits start blank.

ID	Exact product / core / lot	L (in)	f (Hz)	UW (lb/in)	Maker data ID
S1					
S2					
S3					
S4					
S5					

Bounds and positions

ID	dL (in)	df (Hz)	dUW fraction	x from axis (in)	c from face (in)	Bounds / photo ID
S1						
S2						
S3						
S4						
S5						

$T_{nom} = UW (2 L f)^2 / 386.4$. $T_{upper} = UW (1+dUW) [2 (L+dL) (f+df)]^2 / 386.4$. Each bound must have a basis; a blank is not zero. $y = d/2+c$ is measured from the actual spine axis.

Reviewed working limits - all limits for the exact configuration

ID	String limit (lbf)	Tuner limit (lbf)	End/anchor limit (lbf)	Review / source ID
S1				
S2				
S3				
S4				
S5				

Per-string limit = lower of 20 lbf planning ceiling and the three reviewed limits. Include both ends, safe pitch/scale range, raw rating kind/value and all factors once. Reviewer/date: _____
L equality within 1/16 in unless intentionally scaled: _____ String spacing 1.5-2.25 in: _____ Face clearance ≥ 1.5 in: _____
 T_{upper} total: _____ lbf.

03 / Bars, contact and support geometry

Companion tab: Joint Inputs. Record upper and lower assemblies separately. Material, dimensions, orientation, support/contact and positive vertical load path must agree with the same drawing.

Field	Measurement / selected input	Unit	Upper	Lower	Evidence IDs
J01	Selected architecture	text			
J02	Joint part / drawing revision	text			
J03	Bar material, alloy, temper / grade	text			
J04	Bar material lot / certificate ID	text			
J05	Bar total length Lx	in			
J06	Bar depth By toward y	in			
J07	Bar vertical height Hz	in			
J08	Left effective support edge x	in			
J09	Right effective support edge x	in			
J10	Joint reference x	in			
J11	Joint reference y	in			
J12	Joint reference z	in			
J21	Backing plate width across x	in			
J22	Backing plate height along z	in			
J23	Backing plate thickness along y	in			
J24	Backing plate material/lot	text			
J33	Reviewed tension-to-compression lever h_eff	in			
J34	Reviewed front/back contact lever a_eff	in			
J35	Effective perpendicular-grain contact area	in ²			
J36	Positive vertical stop / shoulder / pin detail ID	text			
J37	Tuner/head-block or hitch/break-angle detail ID	text			
J38	Reviewed bar net Iy	in ⁴			
J39	Reviewed bar net Sy	in ³			
J40	Bar net-section derivation reference	text			
J41	Reviewed bar elastic modulus E	psi			

By is depth toward the strings (y); Hz is vertical height (z). Under vertical loads an x-spanning rectangular bar has $I_y = B_y H_z^3/12$ and $S_y = B_y H_z^2/6$. Actual net properties must include holes. Do not credit a collar with vertical capacity from friction alone.

04 / Bolts and force-transfer details

Companion tab: Joint Inputs. Specify the actual grade-marked bolt lot, thread, grip, washers, nuts and shear planes. A nominal diameter is not a fastener rating.

Field	Measurement / selected input	Unit	Upper	Lower	Evidence IDs
J13	Actual bolt count	count			
J14	Bolt shank diameter D	in			
J15	Finished bolt hole diameter Dh	in			
J16	Bolt specification / lot / supplier	text			
J17	Bolt grade markings / photo	text			
J18	Shank grip length	in			
J19	Verified tensile stress area At	in^2			
J20	Threads in shear plane / grip-stack detail	text			
J25	Washer outside diameter	in			
J26	Washer inside diameter	in			
J27	Washer thickness	in			
J28	Nut retention / tightening specification	text			
J29	Bolt 1 x coordinate	in			
J30	Bolt 1 z coordinate	in			
J31	Bolt 2 x coordinate	in			
J32	Bolt 2 z coordinate	in			
J42	Effective shear-sharing bolt count n_s	count			
J43	Effective tension-row bolt count n_t	count			
J44	Additional prying force Q on tension row	lbf			

Required attached fastener/wood detail

For each bolt: x/y/z and diameter of each wood/metal hole; loaded and unloaded edges; end and group spacing along/across grain; main/side member thickness; contact and backing-plate footprint; bolt bending-yield and steel strength data; threads in each shear plane; tightening and retention method.

The workbook reports P/n_s and $(|Mx|/h_{eff}+Q)/n_t$ as screens only. h_{eff} is the actual tension/compression-resultant lever. Q is additional row prying force from the joint calculation. Neither bolt count nor bolt spacing alone determines the distribution.

Drawing / fastener / grain-direction evidence IDs: _____

05 / Hole map and fit checks

Companion tab: Joint Inputs. Coordinates use the marked common axes. Retain a dimensioned drawing of every hole, counterbore, bushing, support edge and contact surface.

Point	x (in)	y (in)	z (in)	Hole D (in)	Evidence / drawing ID
U-S1					
U-S2					
U-S3					
U-S4					
U-S5					
L-S1					
L-S2					
L-S3					
L-S4					
L-S5					

Field	Measurement / selected input	Unit	Upper	Lower	Evidence IDs
J45	Required left end-hole center allowance	in			
J46	Required right end-hole center allowance	in			
J47	Required top bolt-center allowance on pierced plate	in			
J48	Required bottom bolt-center allowance on pierced plate	in			
J49	Actual top center-to-edge bolt allowance	in			
J50	Actual bottom center-to-edge bolt allowance	in			
J51	All wood/metal end, edge and spacing checks reviewed	Y/N			
J52	Free-body/contact/capacity calculation ID	text			
J53	Both bolt rows pass through this same bar	Y/N			
J54	Bar center x coordinate	in			
J55	Simple bar point-load model confirmed	Y/N			

Bar end allowances are hole-center-to-end distances, selected from the actual material/hole/load review. For two rows through the same bar, Hz must contain |z2-z1| plus both positive vertical center allowances. A 1.5 in-high bar cannot contain 1.5 in bolt spacing with positive top and bottom margins. An extended head block/yoke requires its own detail.

06 / Spine calculation inputs

Companion tabs: Stock Inputs, Load Cases and Spine Calc. Reviewer-selected material and restraint properties belong here. They are not inferred from clear-wood ultimate averages.

Field	Measurement / input	Value	Unit	Evidence / bounds / method
M14	Effective length factor Kx		ratio	
M15	Effective length factor Ky		ratio	
M16	Reviewed elastic modulus E for deflection		psi	
M17	Reviewed stability modulus E_min		psi	
M24	Parallel speaking-span force model confirmed		YES/NO	
M25	Same x and face gap c at both speaking ends		YES/NO	
M26	Complete upper/lower free-body drawing ID		text	
M27	Critical cuts and hole geometry calculation ID		text	
M28	Long-term, dynamic and retuning case plan ID		text	
M29	Net section x/y axes principal and section model reviewed		YES/NO	
M31	Upper net area at critical cut		in^2	
M32	Upper net Sx (governing extreme fiber)		in^3	
M33	Upper net Sy (governing extreme fiber)		in^3	
M34	Upper net centroid x offset from gross axis		in	
M35	Upper net centroid y offset from gross axis		in	
M36	Lower net area at critical cut		in^2	
M37	Lower net Sx (governing extreme fiber)		in^3	
M38	Lower net Sy (governing extreme fiber)		in^3	
M39	Lower net centroid x offset from gross axis		in	
M40	Lower net centroid y offset from gross axis		in	
M41	Net-section derivation/review reference		text	

Calculation record

$A=b\,d$; $I_x=b\,d^3/12$; $I_y=d\,b^3/12$; $S_x=b\,d^2/6$; $S_y=d\,b^2/6$. For principal axes: stress = $P/A \pm M_x/S_x \pm M_y/S_y$. Translate moments to each net centroid before using its section properties.

Euler screen: $P_{cr}=\pi^2\,E_{min}\,I/(K\,L)^2$, separately for both axes. It is not an allowable column load. Verify actual unsupported lengths, end restraint, bow, member interaction, second-order response, long-term deflection and clearance under the adopted method.

07 / Calculated demand record

Companion tabs: Load Cases, Spine Calc and Joint Calc. Write the governing case and reference with each result. Component maxima may occur in different cases; they are not automatically a simultaneous load vector.

Quantity	Unit	Upper / value	Lower / value	Governing case / calculation
Total upper-bounded string tension	lbf			
Gross spine compression envelope	psi			
Gross spine tension envelope	psi			
Upper critical net compression / tension	psi			
Lower critical net compression / tension	psi			
Euler Pcr, x / y (screen)	lbf			
Joint P max at each end	lbf			
Joint Mx max at actual datum	lbf in			
Joint My max at actual datum	lbf in			
Left bar support-root moment	lbf in			
Right bar support-root moment	lbf in			
Bar gross I_y / S_y	in^4 / in^3			
Bar net S_y at critical section	in^3			
Bar net bending-stress screen	psi			
Mx-only tension-row force before Q	lbf			
Mx-only per-bolt force including Q	lbf			
Equal-share vertical shear/bolt screen	lbf			
Collar face-couple force	lbf			
Collar effective contact pressure	psi			
Minimum left / right end allowance	in			
Same-bar vertical fit	result			
Predicted bar/frame deflection & joint slip	in			
Remaining vessel clearance at reviewed loads	in			

Static envelope: all 32 combinations of each string at 0 or T_{upper} . For parallel strings, $P = \sum T_i$, $M_x = \sum T_i y_i$, $M_y = -\sum T_i x_i$. For an actual joint datum (x_j, y_j) , $M_{x_j} = M_x - P y_j$ and $M_{y_j} = M_y + P x_j$. Local bends, tuner wraps, impact, fatigue and creep need their own reviewed calculations.

08 / Demand-to-limit review register

Companion tab: Review Limits. Enter an adjusted allowable on the same load/design basis as the demand. For an interaction index, use its cited permitted index. No numeric capacity is supplied by this form.

ID / check	Unit	Demand D	Allowable C	D/C	Case / review calculation ID
SP01 Spine gross axial + biaxial member interaction	ratio				
SP02 Spine upper critical net section	ratio				
SP03 Spine lower critical net section	ratio				
SP04 Spine stability / buckling review	ratio				
SP05 Frame deflection / long-term clearance loss	in				
U01 Bar net elastic bending stress	psi				
U02 Bar deflection / clearance allowance	in				
U03 Metal bearing / tear-out / block shear	ratio				
U04 Bolt/wood lateral yield-mode connection	lbf				
U05 Bolt tension + shear + prying interaction	ratio				
U06 Backing plate / washer / wood contact	ratio				
U07 Wood splitting / group / local net failures	ratio				
U08 Positive vertical stop / restraint path	lbf				
U09 Tuner / head-block / local contact path	ratio				
L01 Bar net elastic bending stress	psi				
L02 Bar deflection / clearance allowance	in				
L03 Metal bearing / tear-out / block shear	ratio				
L04 Bolt/wood lateral yield-mode connection	lbf				
L05 Bolt tension + shear + prying interaction	ratio				
L06 Backing plate / washer / wood contact	ratio				
L07 Wood splitting / group / local net failures	ratio				
L08 Positive vertical stop / restraint path	lbf				
L09 Hitch / bridge / base-shoe local path	ratio				

Each record also needs raw rating kind/value (WLL, allowable, proof, breaking or yield), every adjustment/design factor applied once, governing failure mode, actual material/geometry basis, reviewer and review date. Put the full derivation in the workbook or attached calculation. N/A requires a dated technical justification.

Global reviewer / date: _____ Calculation package ID: _____

09 / Exact Rev B row and closeout

Copy the final row back only with configuration-specific values and dated review evidence. A complete worksheet prepares the separate release review; it does not itself release fabrication or tensioning.

Bar material and calculated connection limits

Required record	Upper joint	Lower joint
Actual bar material / grade / lot		
Bar Lx x By x Hz and orientation		
Bolt, back plate, contact/positive-stop detail ID		
Reviewed Vz allowable (lbf)		
Reviewed Mx / My allowables (lbf in)		
Combined-load domain / other force components		
Controlling failure mode and load case		
Calculation / source / drawing revision IDs		
Reviewer and review date		

Retain every other original worksheet row

Original Rev B record	Value / evidence ID	State
Vessel ID profile and mouth at each height		OPEN
Actual spine section and orientation		OPEN
String plane and outer-string coordinates		OPEN
Anchor hole size, spacing, end allowance		OPEN
Frame joints and operating-load limits		OPEN
Mass and center of gravity in three axes		OPEN
Foot contacts and stability margin		OPEN
Restraint geometry and rated load basis		OPEN
Sensor mounting, excitation, and SPL ceiling		OPEN

Actual configuration ID: _____ Reviewed total operating limit: _____ lbf
Limit-source ID: _____ Package reviewer / date: _____

CURRENT STATE: OPEN / incomplete actual measurements and review. FABRICATION / TENSIONING: NOT RELEASED. Unknown geometry, ratings, load paths or unreviewed prerequisites leave the strings slack.

10 / Source corrections and references

The source documents remain unchanged. These corrections prevent inconsistent examples from becoming real-build inputs. Full traceable notes and URLs are on EX-B1 Audit and Sources.

A01 | G3 is printed as 195 Hz, but A4=440 equal temperament gives 195.997718 Hz. The source UW values produce totals of 70.702909 lbf using printed pitches, or 70.835406 lbf using the stated tuning convention. Actual pitch choice is still unfilled.

A02 | With a 1.5 in-wide support and $x=-4,-2$ in loads, left root $M = 16.25(3.25)+13.02(1.25)=69.0875$ lbf in. The source 113 lbf in is not reproduced by those stated assumptions.

A03-A05 | Rectangle $S/A=d/6$, not $d/2$. Speaking length is not automatically effective buckling length. For the ideal centered 0.375 in through-hole in a 1.5 x 2.5 in member, worst-cut $A=2.8125$ in² and $S_x=1.171875$ in³; local hole effects remain separate.

A06-A07 | M/h is a declared force couple, not each bolt's prying force. Bolt spacing along z must fit the actual pierced plate/head block; it is not limited by the spine depth y . CX-1 needs a consistent dimensioned joint.

A08-A10 | Declare bar B_y and H_z before calculating bending. Select end margins from actual material/hole/load checks. Clear-wood ultimates, approximate bolt capacities and a supposed failure ranking do not establish a rating. Do not discard the string-plane couple.

A11-A12 | The current maker tension formula was verified; the legacy UW booklet was not retrieved. EX-B1 UWs remain inherited example data. Static screens and a 30-minute observation do not establish impact, creep, fatigue or proof-load capacity.

Project lineage

P1. Rev B treats string tension as a --gate--, not a decoration. Drive revision read: 2026-09-08 14:56:53 UTC.
<https://docs.google.com/document/d/1SIbBJlrwtkJfYzcllevJsoiEVgszdl-GTHP6-9ICJ0/edit>

P1a. Five_String_Acoustic_Resonator_Rev_B_Schematic_Update(1).docx, Rev B dated 5 September 2026; pages 4-6. Page 5 contains the original measurement worksheet and exact connection-limits row.

Primary technical references

P2. D'Addario, All About Guitar String Tension - maker equation.
<https://www.daddario.com/blogs/guitar/all-about-string-tension>

P3. D'Addario String Tension Pro - obtain exact product/scale/pitch evidence.
<https://www.daddario.com/pages/string-tension-pro-string-tension-calculator>

P4-P6. USDA FPL-GTR-282 (2021): chapter 7, design properties; chapter 8, fastenings; chapter 9, structural analysis.
<https://research.fs.usda.gov/treesearch/62252> | [/62253](https://research.fs.usda.gov/treesearch/62253) | [/62251](https://research.fs.usda.gov/treesearch/62251)

P7. American Wood Council, 2024 NDS - choose the applicable edition/provisions with the reviewer.
<https://awc.org/resources/2024-nds/>

No numeric wood, metal, bolt or joint allowable is adopted from these reference entry points. Workbook formulas calculate demands under declared models; actual capacities and release decisions require their separate documented review.